Data Structures and Algorithms

INFORMATION
TECHNOLOGY

Objectives

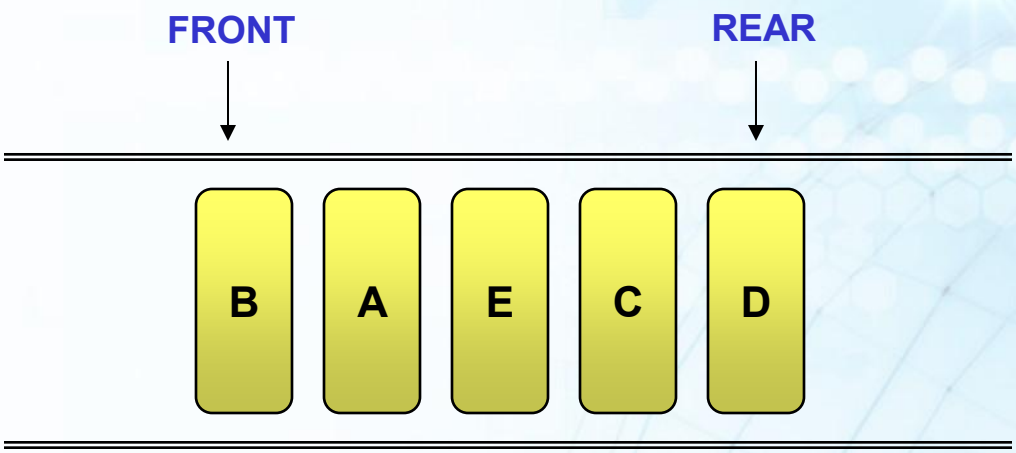
- In this session, you will learn to:
  - Solve programming problems by using queues
  - Store data in a tree

Solving Programming Problems by Using Queues

- Consider a situation where you have to create an application with the following set of requirements:
  - The application should serve the requests of multiple users.
  - At a time, only one request can be processed.
  - The request, which came first should be given priority.
- However, the rate at which the requests are received is much faster than the rate at which they are processed.
- How can you solve this problem?
  - You can solve this problem by storing the requests in such a manner so that they are retrieved in the order of their arrival.
  - A data structure called queue stores and retrieves data in the order of its arrival.
  - A queue is also called a FIFO list.

Defining Queues

- Queue is a list of elements in which an element is inserted at one end and deleted from the other end of the queue.
- Elements are inserted at the rear end and deleted from the front end.



Just a Minute

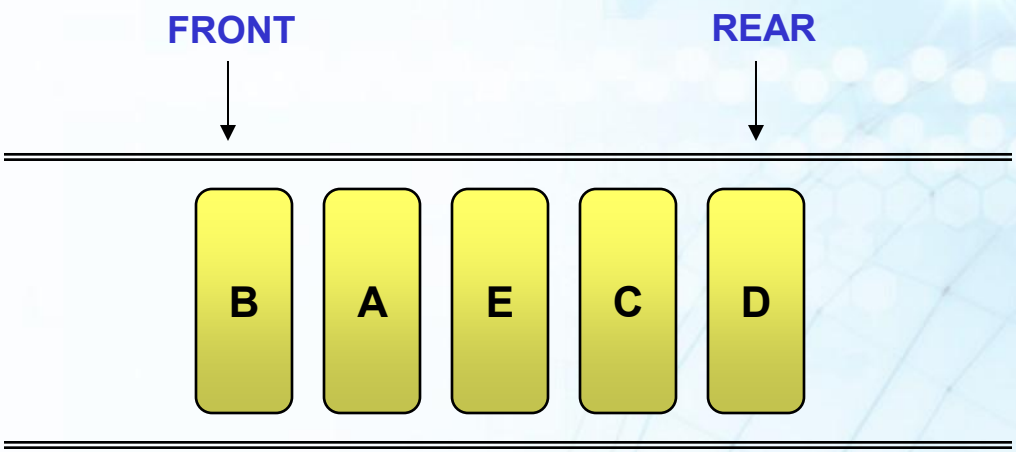
- A queue is also known as a _____ list.

Just a Minute (Contd.)

- Solution:
  - FIFO

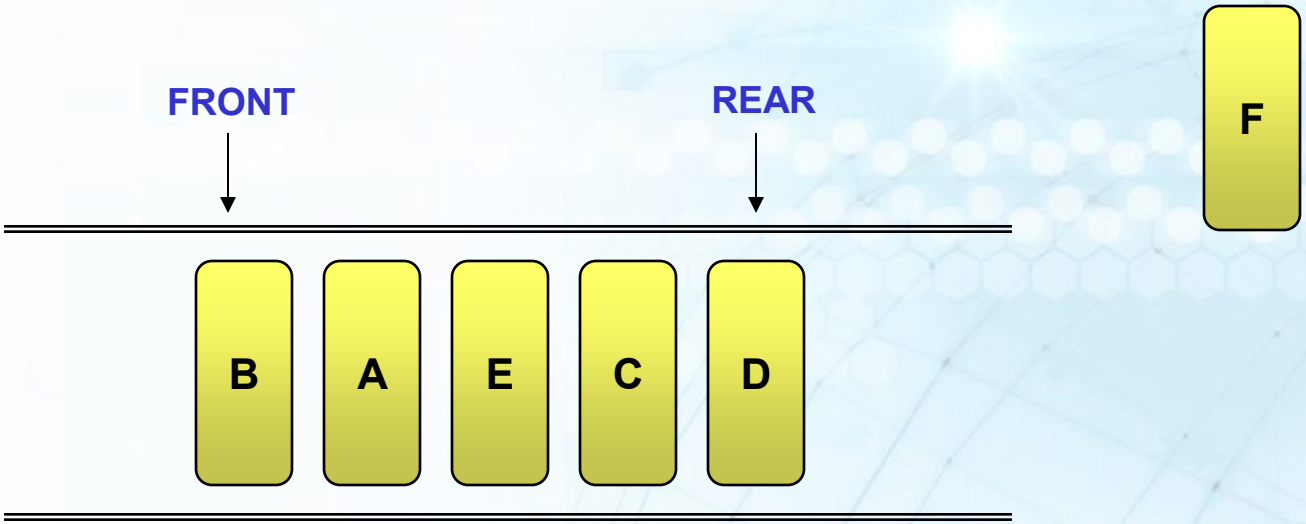
Identifying the Various Operations on Queues

- Various operations implemented on a queue are:
  - Insert
  - Delete



Identifying the Various Operations on Queues (Contd.)

- Insert:
  - It refers to the addition of an item in the queue.
  - Suppose you want to add an item F in the following queue.
  - Since the items are inserted at the rear end, therefore, F is inserted after D.
  - Now F becomes the rear end.



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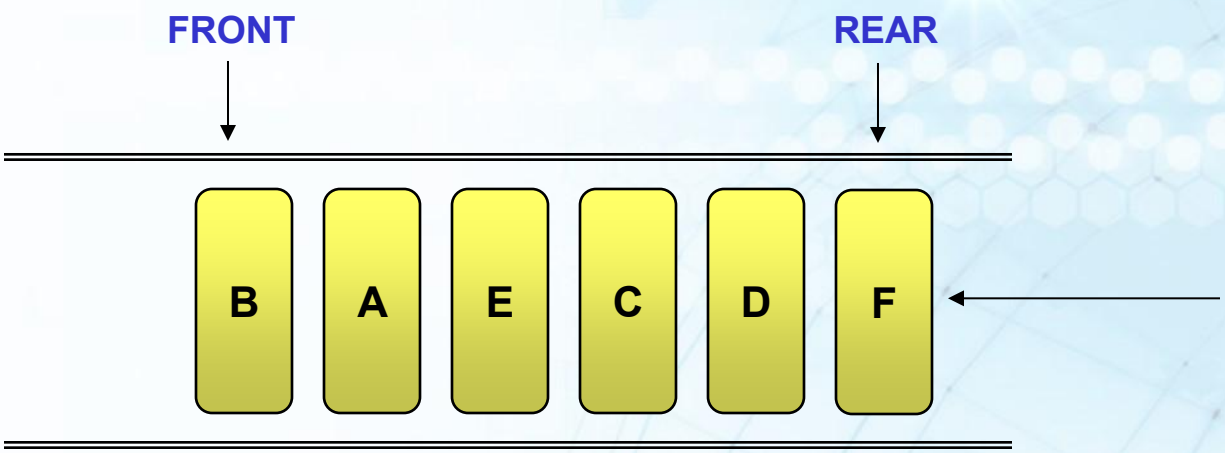
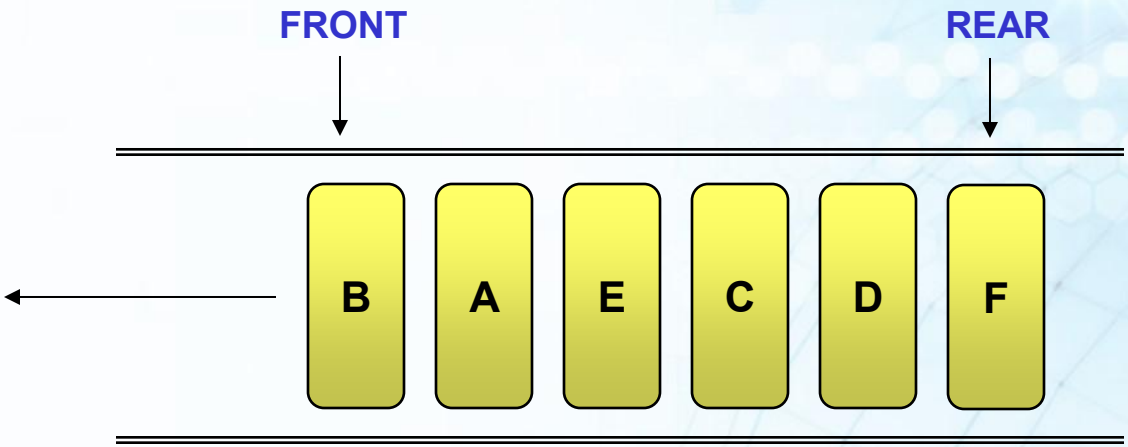


Identifying the Various Operations on Queues (Contd.)

- Delete:
  - It refers to the deletion of an item from the queue.
  - Since the items are deleted from the front end, therefore, item B is removed from the queue.
  - Now A becomes the front end of the queue.



Identifying the Various Operations on Queues (Contd.)

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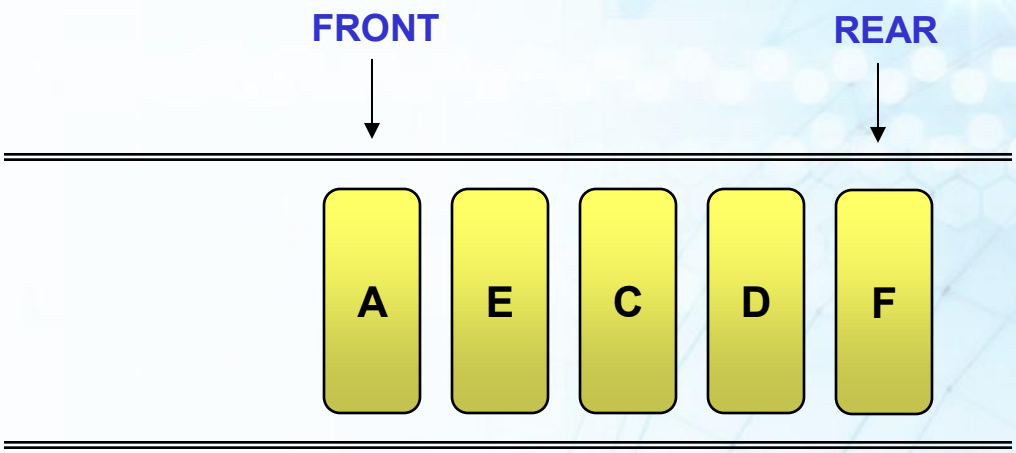


Just a Minute

- Queues are data structures in which data can be added at one end called _____.

Just a Minute (Contd.)

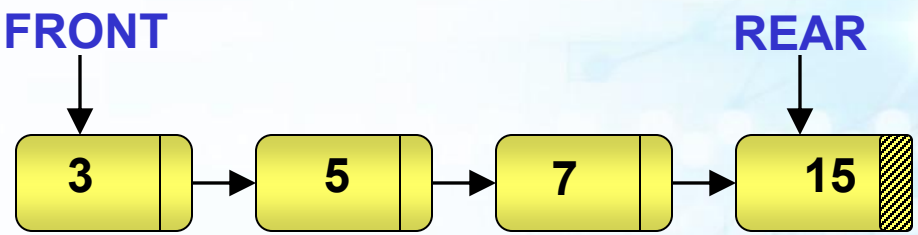
- Solution:
  - Rear

Implementing a Queue

- You can implement a queue by using an array or a linked list.
- A queue implemented in the form of a linked list is known as a linked queue.

Implementing a Queue (Contd.)

- To keep track of the rear and front positions, you need to declare two variables/pointers, `REAR` and `FRONT`.
- If the queue is empty, `REAR` and `FRONT` point to `NULL`.



Let us view an animation to understand the insertion of an element in a queue.

Demo: Solving Programming Problems by Using Queues

- Problem statement:
  - Write a program to implement insert and delete operations on a linked queue.

Demo: Solving Programming Problems by Using Queues (Contd.)

- Solution:
  - To implement insert and delete operations on a linked queue, you need to perform the following tasks:
    1. Write the code.
    2. Execute the code and verify the output.

Storing Data in a Tree

- Consider a scenario where you are required to represent the directory structure of your operating system.
- The directory structure contains various folders and files. A folder may further contain any number of sub folders and files.
- In such a case, it is not possible to represent the structure linearly.
- In such a case, it would be good if you have a data structure that enables you to store your data in a nonlinear fashion.

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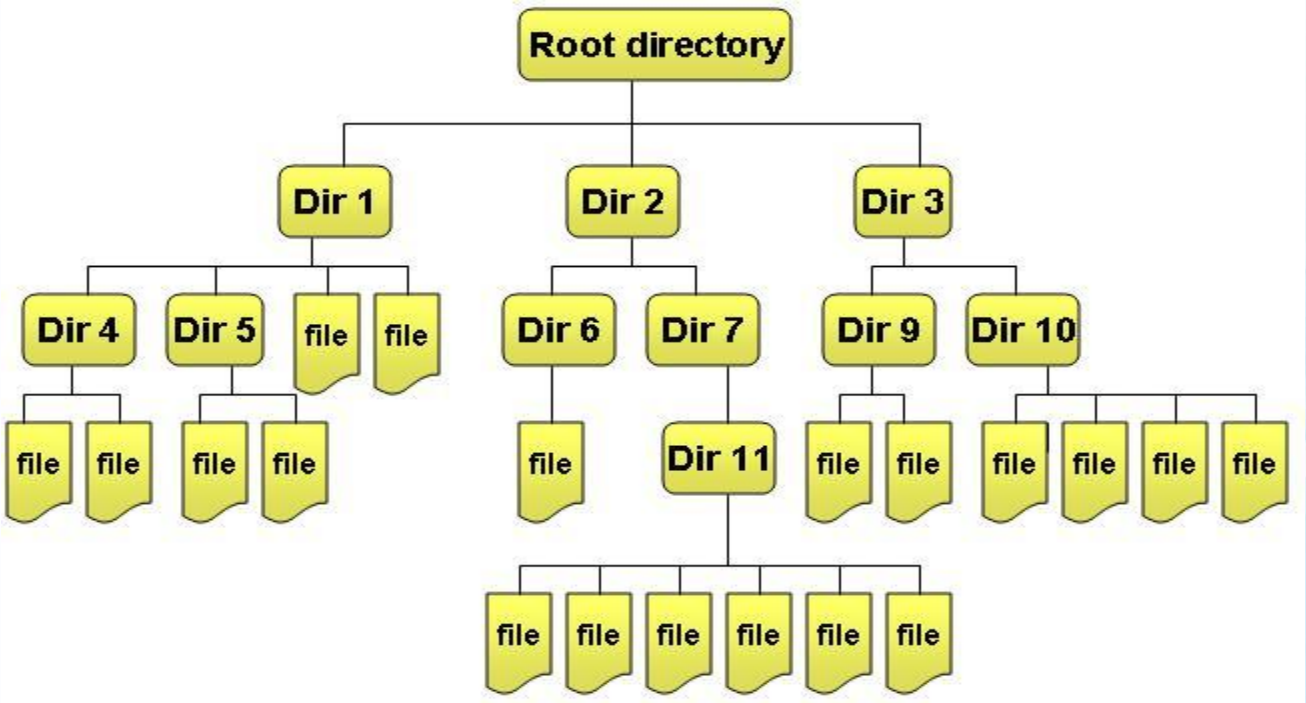
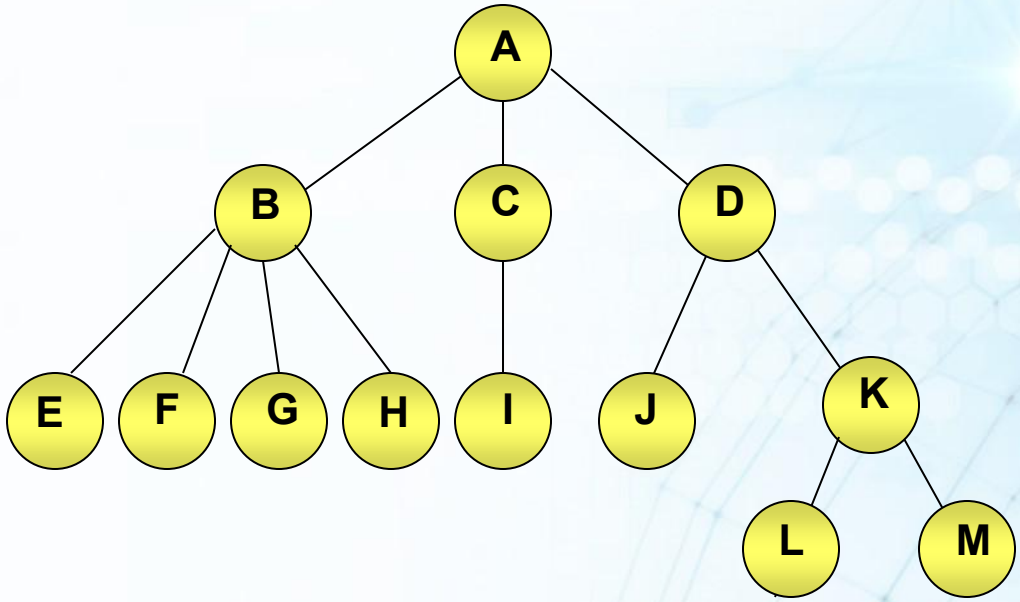
Storing Data in a Tree (Contd.)

- Directory structure:



Defining Trees

- A tree is a nonlinear data structure that represent a hierarchical relationship among the various data elements.
- Trees are used in applications in which the relation between data elements needs to be represented in a hierarchy.



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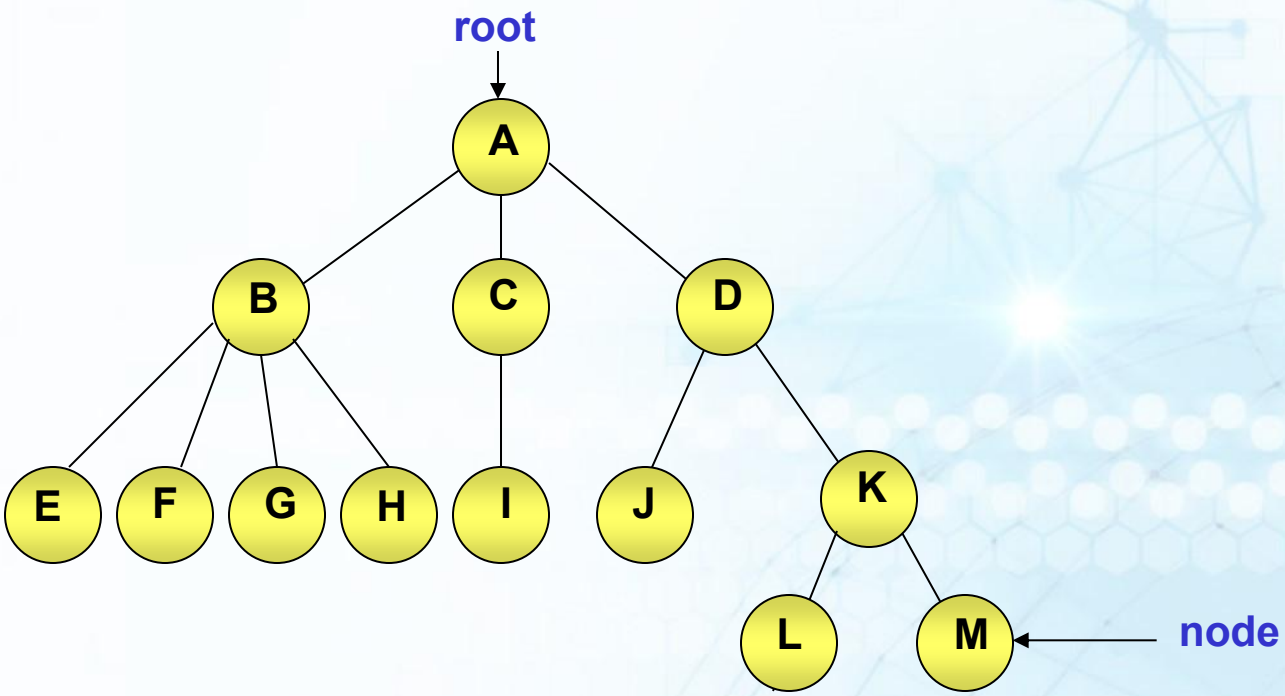
Defining Trees (Contd.)

- The topmost node in a tree is called root.



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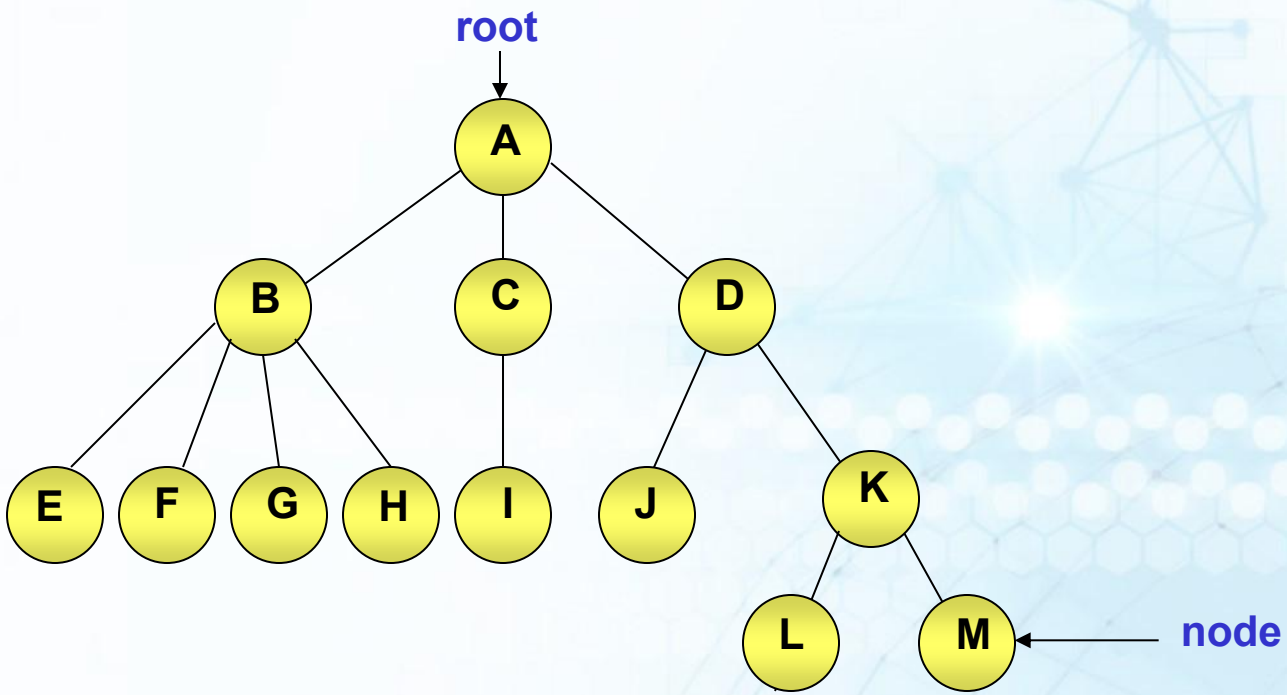
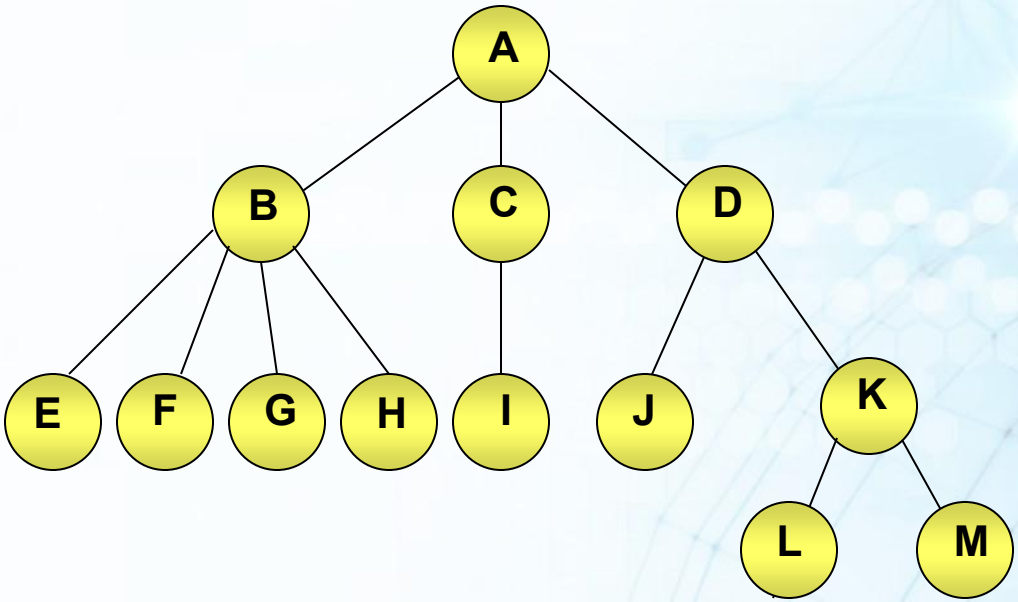
Defining Trees (Contd.)

- Each node in a tree can further have subtrees under its hierarchy.



Tree Terminology

- Leaf node: It refers to a node with no children.



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Tree Terminology (Contd.)

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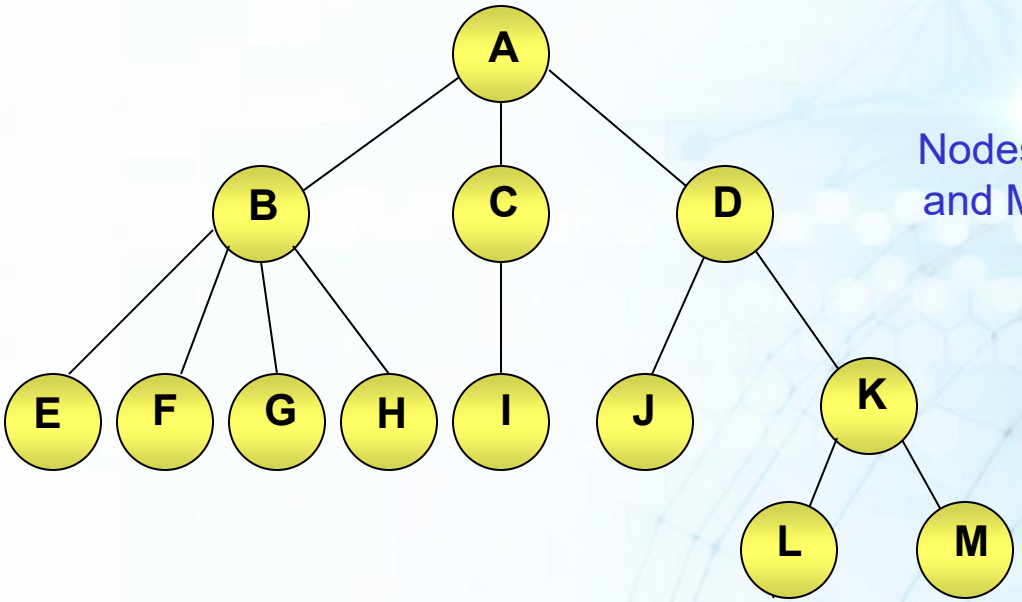

Nodes E, F, G, H, I, J, L, and M are leaf nodes.

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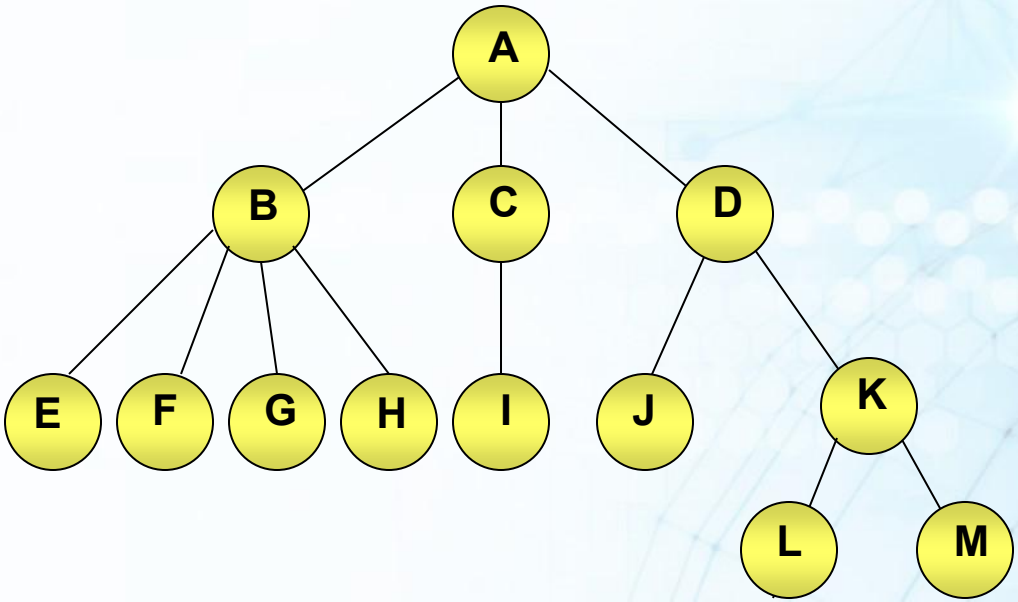
Tree Terminology (Contd.)

- Subtree: A portion of a tree, which can be viewed as a separate tree in itself is called a subtree.



Data Structures and Algorithms

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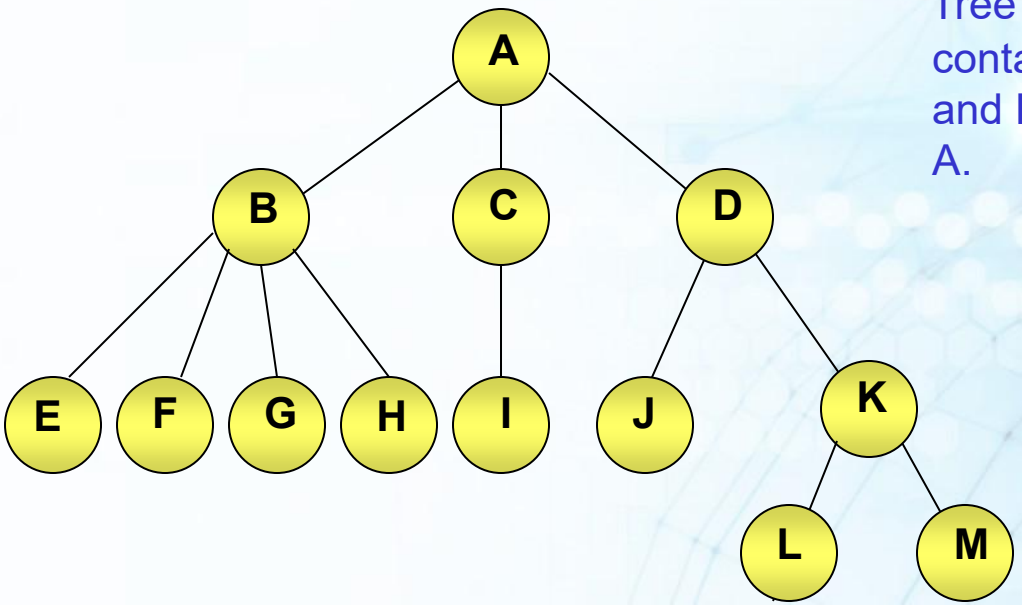
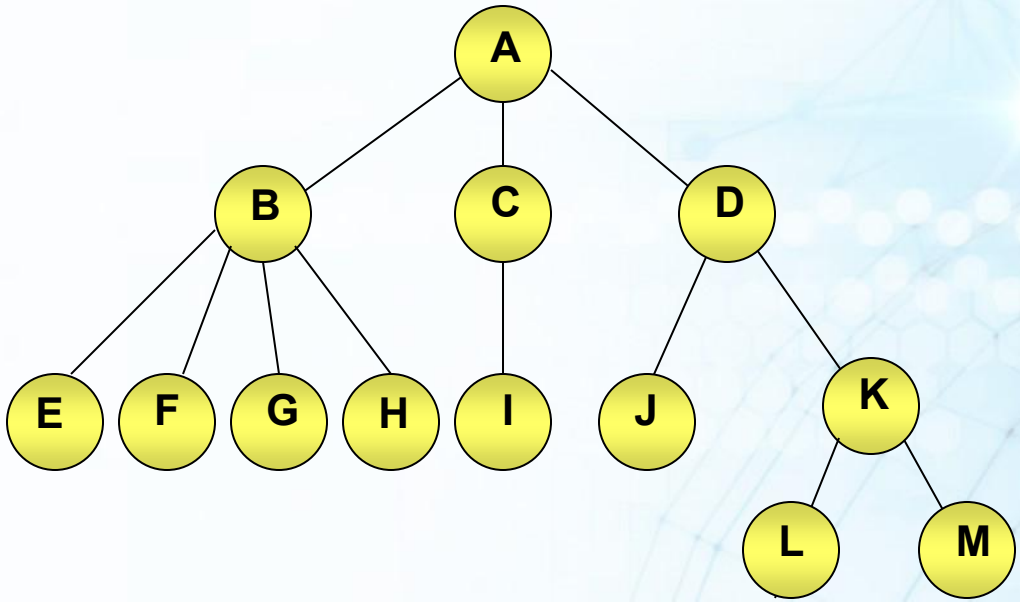


Tree with root B, containing nodes E, F, G, and H is a subtree of node A.

Tree Terminology (Contd.)

- Children of a node: The roots of the subtrees of a node are called the children of the node.



Data Structures and Algorithms

Tree Terminology (Contd.)

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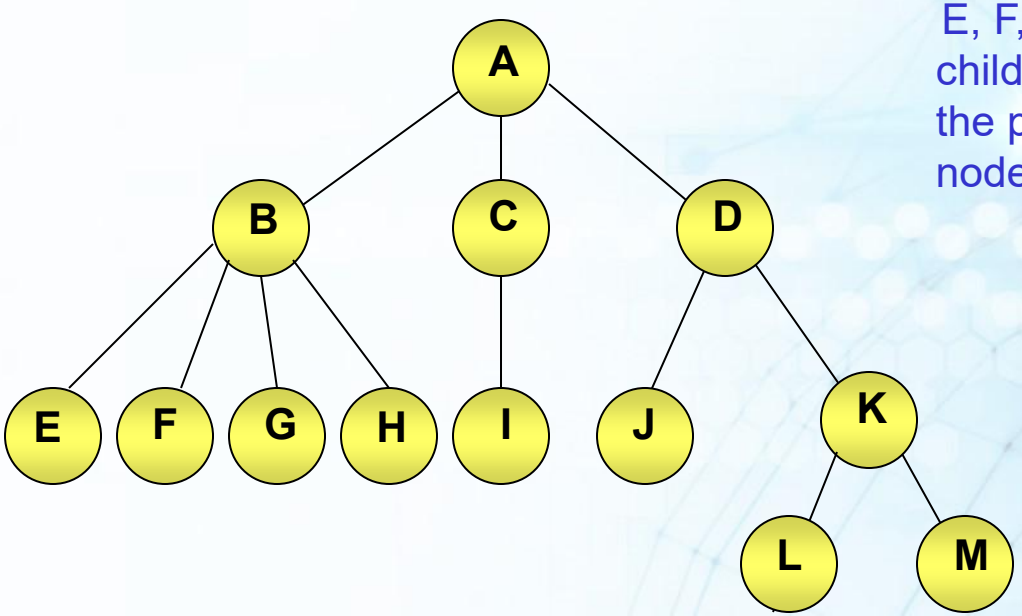

E, F, G, and H are children of node B. B is the parent of these nodes.

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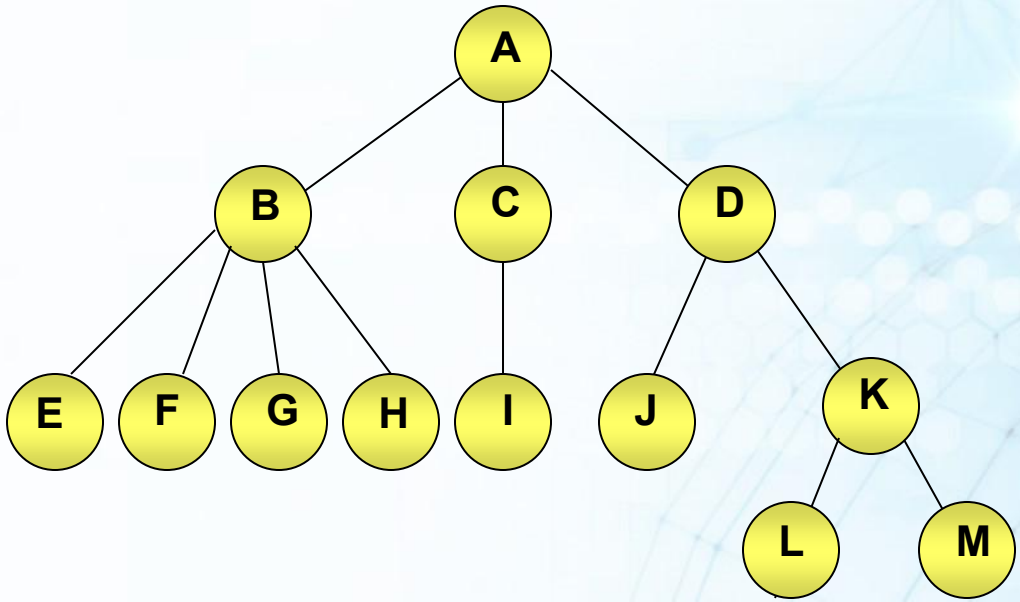
Tree Terminology (Contd.)

- Degree of a node: It refers to the number of subtrees of a node in a tree.



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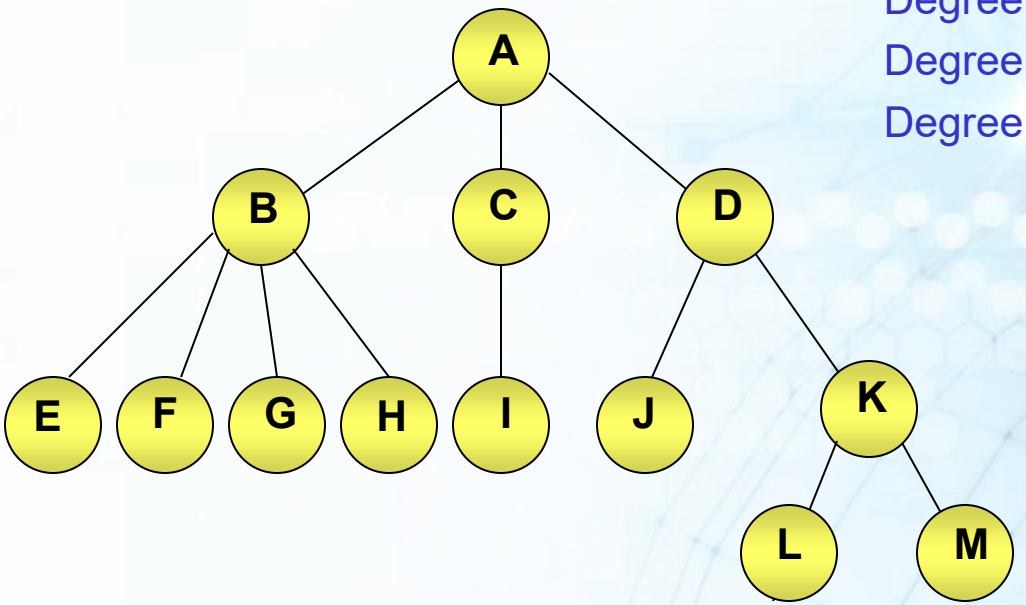
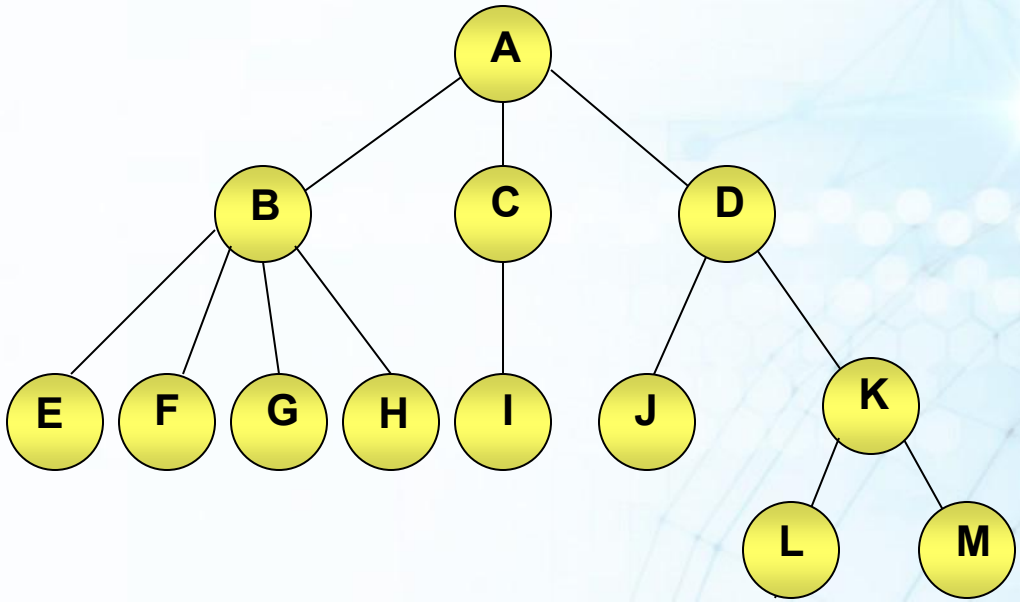


Degree of node C is 1.
Degree of node D is 2.
Degree of node A is 3.
Degree of node B is 4.

Data Structures and Algorithms

Tree Terminology (Contd.)

- Edge: A link from the parent to a child node is referred to as an edge.



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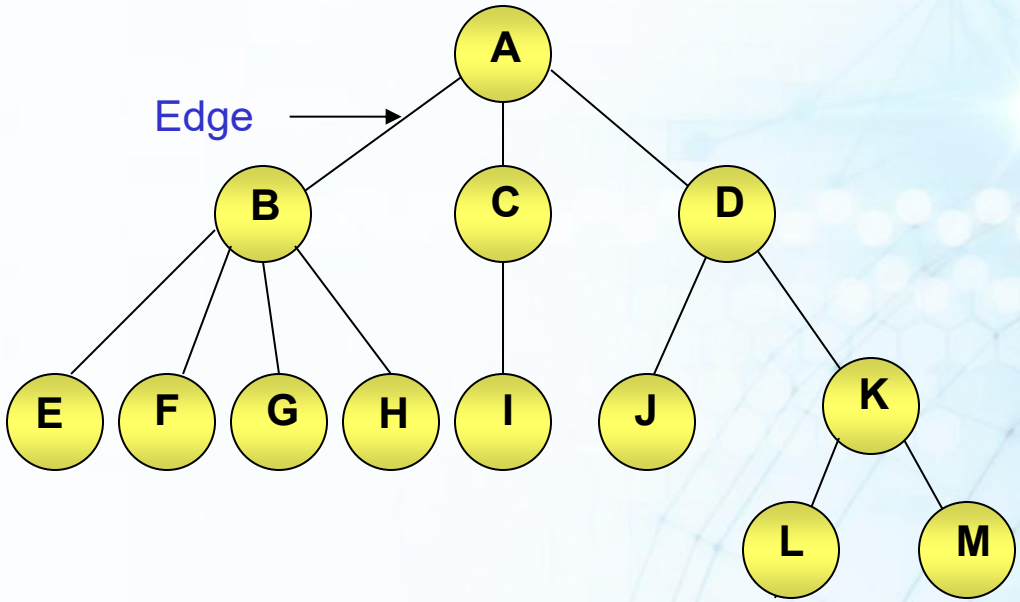
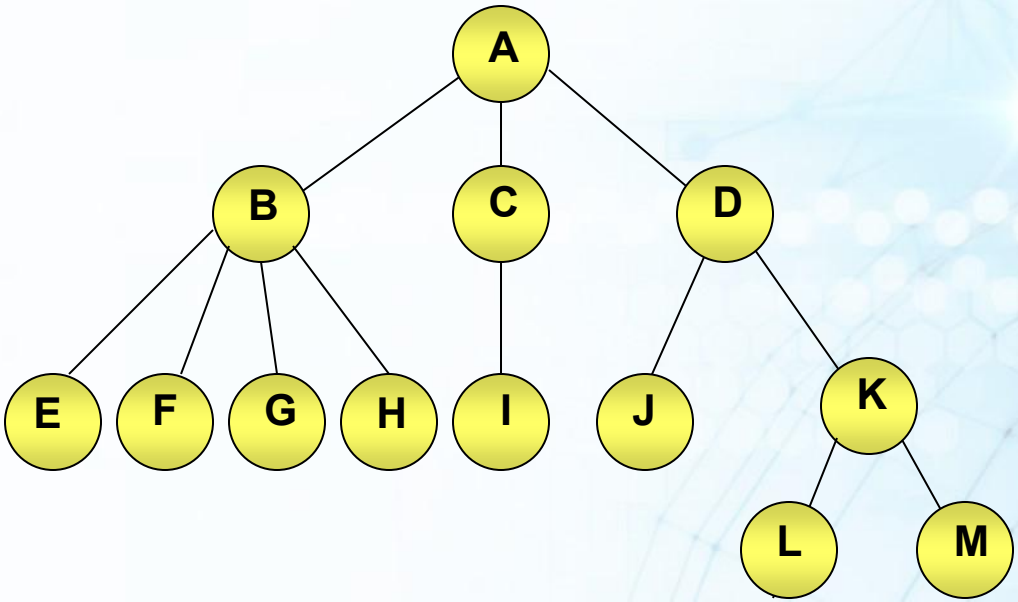


Data Structures and Algorithms

Tree Terminology (Contd.)

- Siblings/Brothers: It refers to the children of the same node.



Data Structures and Algorithms

Tree Terminology (Contd.)

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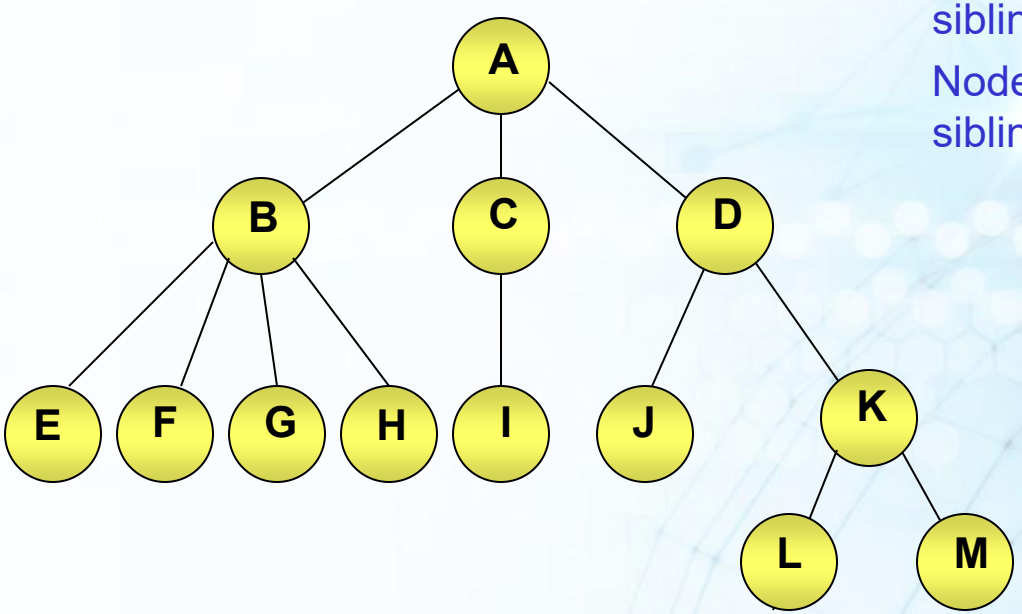


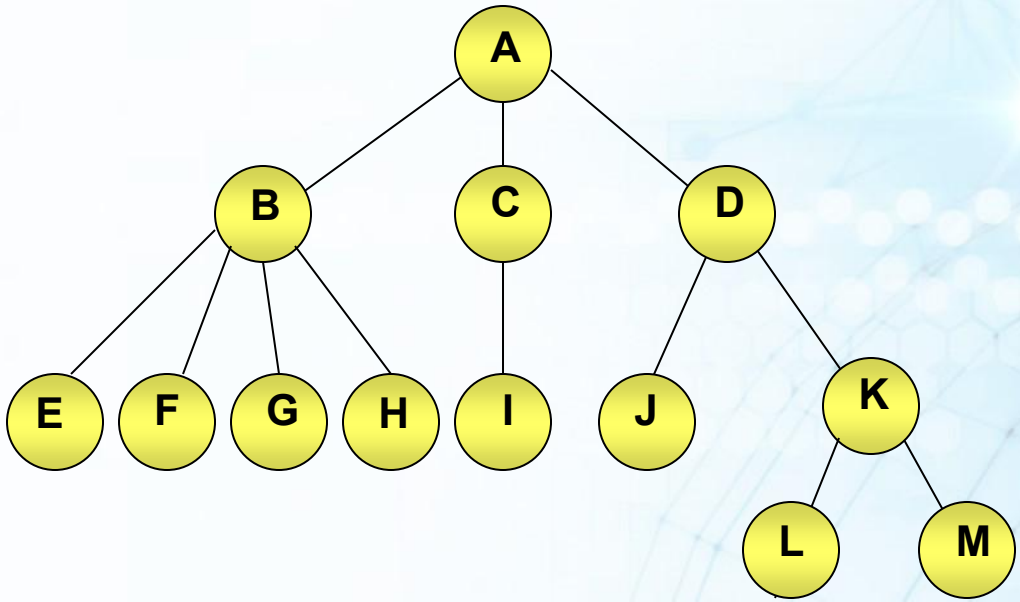
Nodes B, C, and D are siblings of each other.

Nodes E, F, G, and H are siblings of each other.

Data Structures and Algorithms

Tree Terminology (Contd.)

- Internal node: It refers to any node between the root and a leaf node.



Data Structures and Algorithms

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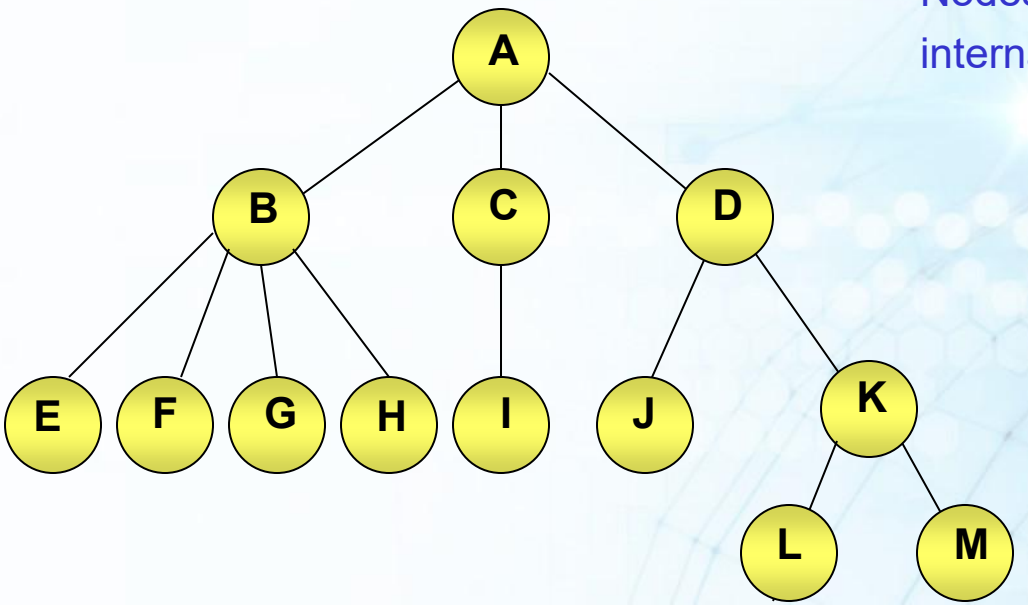
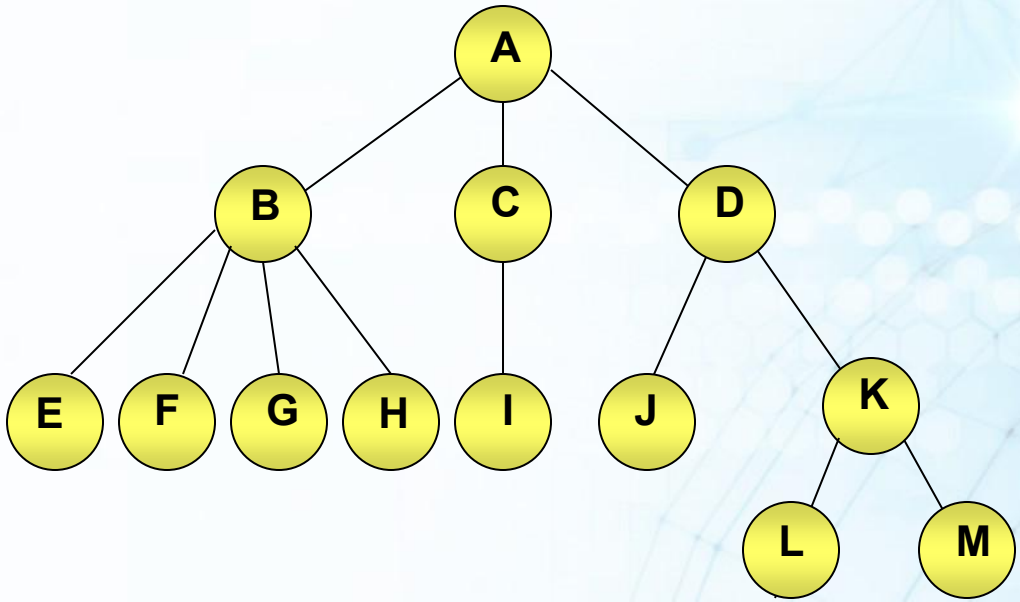


Nodes B, C, D, and K are internal nodes.

Data Structures and Algorithms

Tree Terminology (Contd.)

- Level of a node: It refers to the distance (in number of nodes) of a node from the root.



Data Structures and Algorithms

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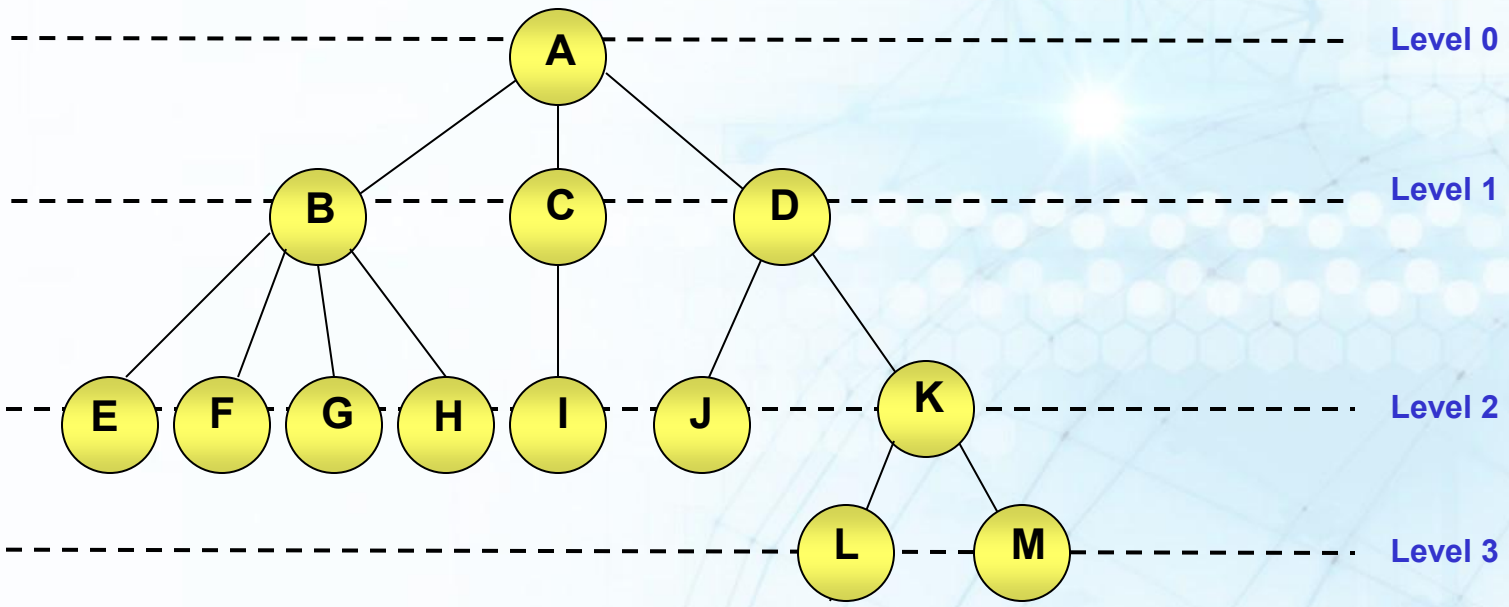

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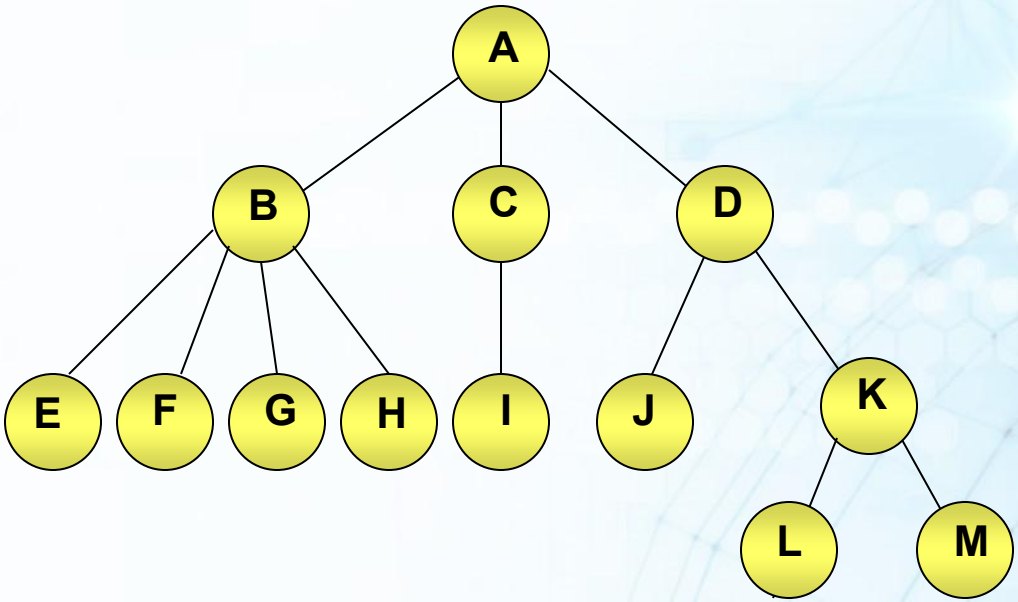
Tree Terminology (Contd.)

- Depth of a tree: Refers to the total number of levels in the tree. The depth of the following tree is 4.



Just a Minute

- A node without any children is called a _____ node.

Just a Minute (Contd.)

- Solution:
  - Leaf

Summary

- In this session, you learned that:
  - A queue is a list of elements in which items are inserted at one end of the queue and deleted from the other end of the queue.
  - The end at which elements are inserted is called the rear, and the end from which the elements are deleted is called the front.
  - A queue is called a FIFO data structure.
  - The two types of operations that can be performed on a queue are insert and delete.
  - A queue implemented by using a linked list is called a linked queue.
  - A tree is a nonlinear data structure that represents a hierarchical relationship among the various data elements.